Voltage Sensitivity of Distribution Feeders with Inverter Based Resources

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Abstract—This paper studies how the voltage at the end of a radial distribution feeder responds to active and reactive power injected by inverter based resources. We derive a simple sensitivity expression from the voltage drop formula and validate it on a 33-bus test feeder. The results show that reactive power control remains effective only when the ratio of reactance to resistance is large.

1 Introduction

The voltage regulation problem is examined from a practical point of view in this paragraph number 1. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

The voltage regulation problem is examined from a practical point of view in this paragraph number 2. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

The voltage regulation problem is examined from a practical point of view in this paragraph number 3. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

2 Sensitivity Model

The linearised feeder model is examined from a practical point of view in this paragraph number 1. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

The linearised feeder model is examined from a practical point of view in this paragraph number 2. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

$$\Delta U = (P\,R + Q\,X) / U_2 \quad (1)$$

The sensitivity coefficient is examined from a practical point of view in this paragraph number 1. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

The sensitivity coefficient is examined from a practical point of view in this paragraph number 2. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

Fig. 1. Receiving end voltage versus injected reactive power for three X/R ratios.

3 Case Study

The 33-bus feeder case is examined from a practical point of view in this paragraph number 1. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

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The 33-bus feeder case is examined from a practical point of view in this paragraph number 4. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

Table 1. Parameters of the test feeder and the inverter ratings.

4 Conclusion

The main finding is examined from a practical point of view in this paragraph number 1. We describe how the operating point changes when the network impedance increases, and why the voltage at the receiving end becomes sensitive to reactive power. Such behaviour is well known to operators but rarely quantified for distribution feeders with high penetration of inverter based resources.

References

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